



**UTM**  
UNIVERSITI TEKNOLOGI MALAYSIA

**UNIVERSITI TEKNOLOGI MALAYSIA  
FAKULTI SAINS**

**TEST II  
SEMESTER III**

**COURSE CODE : FSPC 0034**

**COURSE NAME : CHEMISTRY II**

**PROGRAMME : ASASI UTMSPACE**

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**DATE : 27 FEB 2020**

**DURATION : 1.5 HOURS**

**INSTRUCTION TO STUDENTS:**

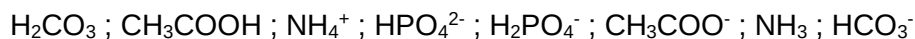
**ANSWER ALL QUESTIONS**

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**QUESTION 1 (30 MARKS)**

1. From the list below, identify the acid/conjugate base pair, or base/conjugate acid pair.



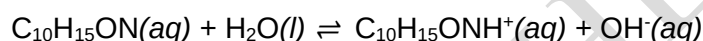
acid :  $\text{H}_2\text{PO}_4^-$ ,  $\text{CH}_3\text{COOH}$ ,  $\text{H}_2\text{CO}_3$       base :  $\text{NH}_3$  (4 marks)  
 c.b. :  $\text{HPO}_4^{2-}$ ,  $\text{CH}_3\text{COO}^-$ ,  $\text{HCO}_3^-$       c.a. :  $\text{NH}_4^+$

2. A mixture of citric acid,  $\text{C}_5\text{H}_7\text{O}_4\text{COOH}$  and sodium citrate can function as a buffer solution. [ $K_a$  for citric acid =  $7.0 \times 10^{-4} \text{ mol L}^{-1}$ ]

- Write a balanced equation to represent the dissociation of citric acid in water.
- A suitable indicator is added when citric acid is titrated by a strong base such as 0.1 M NaOH solution. What is an indicator?

(2 + 1 marks)

3. Ephedrine, a central nervous system stimulant, is used in nasal sprays as decongestant. This compound is a weak organic base:



A 0.035M solution of ephedrine has a pH of 11.33.

- What are the equilibrium concentrations of  $\text{C}_{10}\text{H}_{15}\text{ON}$ ,  $\text{C}_{10}\text{H}_{15}\text{ONH}^+$  and  $\text{OH}^-$
- Calculate the  $K_b$  for ephedrine
- What is the  $K_a$  of  $\text{C}_{10}\text{H}_{15}\text{ONH}^+$

(4 + 2 + 2 marks)

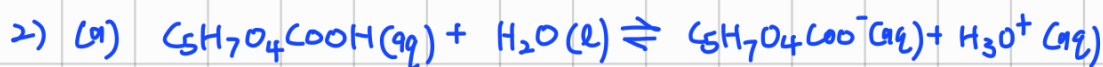
4. A certain hydroiodic acid, HI, ionizes at 0.67 %. Calculate the pH of 0.8 M of the HI. (3 marks)

5. Calculate the pH of  $0.005 \text{ mol dm}^{-3}$  sulphuric acid.

(2 marks)

6. A titration is carried out for 25.00 mL of 0.100 M HCl with 0.100 M NaOH.
- Calculate the volume of NaOH solution needed to reach the equivalence point
  - Calculate the pH at these volumes of added base solution:
    - 0.00 mL
    - 12.50 mL
    - 25.00 mL

(3+3+3+1 marks)



OR



(b) The indicator is a weak acid which changes colour over a specific pH range in solution

$$3) (a) \text{pOH} = 14 - 11.33 \\ = 2.67$$

$$\text{pOH} = -\log([\text{OH}^-])$$

$$[\text{OH}^-] = 10^{-\text{pOH}}$$

$$= 10^{-2.67}$$

$$= 2.13 \times 10^{-3} \text{ M} = x$$



|         |           |    |    |
|---------|-----------|----|----|
| Initial | 0.035     | 0  | 0  |
| Change  | -x        | +x | +x |
| Equil   | 0.035 - x | x  | x  |

$$[\text{OH}^-] = [\text{C}_{10}\text{H}_{15}\text{ONH}^+] = 2.13 \times 10^{-3} \text{ M}$$

$$[\text{C}_{10}\text{H}_{15}\text{ON}] = 0.035 - 2.13 \times 10^{-3} \\ = 0.03287 \text{ M}$$

$$(b) K_b = \frac{(2.13 \times 10^{-3})^2}{0.03287} \\ = 1.38 \times 10^{-4}$$

$$(c) K_w = K_a \times K_b$$

$$K_a = \frac{K_w}{K_b} \text{ (conjugate acid)}$$

$$= \frac{1 \times 10^{-14}}{1.38 \times 10^{-4}}$$

$$= 7.25 \times 10^{-11}$$

$$4) 0.67 = \frac{\text{moles ionised}}{\text{moles available}} \times 100$$

$$0.67 = \frac{\text{moles ionised}}{0.8} \times 100$$

$$\text{moles ionised} = 5.36 \times 10^{-3} = [\text{H}^+] = [\text{I}^-]$$

$$\text{pH} = -\log[\text{H}^+]$$

$$= 2.27$$



$$[\text{H}^+] = 2 \times 0.005$$

$$= 0.01 \text{ M}$$

$$\text{pH} = -\log(0.01)$$

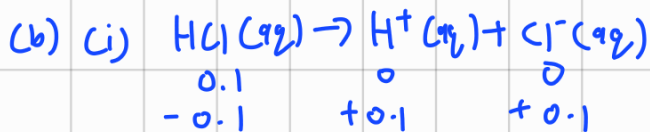
$$= 2$$



$$\frac{M_a V_a}{M_b V_b} = \frac{a}{b}$$

$$\frac{0.1(\text{M})}{0.1(\text{M})} = \frac{1}{1}$$

$$V_b = 25 \text{ mL}$$



$$\text{pH} = -\log[\text{H}^+]$$

$$= -\log(0.1)$$

$$= 1$$

(ii) mol of  $\text{HCl} = 0.025 \text{ L} \times 0.1$

$$= 0.0025 \text{ mol}$$

mol of  $\text{NaOH} = 0.0125 \text{ L} \times 0.1$

$$= 0.00125 \text{ mol (limiting)}$$



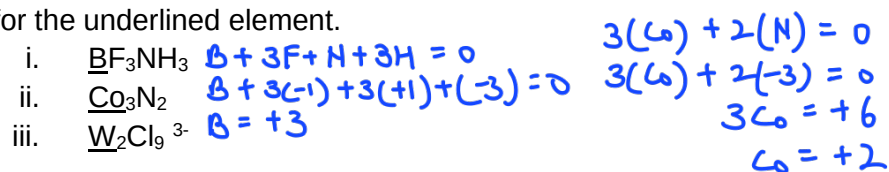
$$\text{pH} = -\log[\text{H}^+] = -\log(0.033) = 1.47$$

(iii) Since the volume and concentrations of the acid and base solutions are the same  
pH 7.0, neutral

$$\begin{aligned}
 2(W) + 9(4) &= -3 \\
 2W + 9(-1) &= -3 \\
 2W &= -3 + 9 \\
 2W &= +6 \\
 W &= +3
 \end{aligned}$$

**QUESTION 2 (20 MARKS)**

1. By applying the rules for calculating oxidation number, determine the oxidation number for the underlined element.



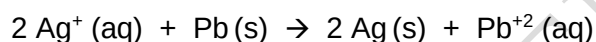
(3 marks)

2. Balance the following reaction in basic solution :



(7 marks)

3. Consider the given overall redox reaction:



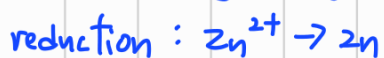
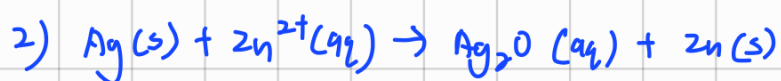
- i. Sketch a voltaic cell with correct labellings  
 ii. Calculate the standard cell potential  
 iii. Write the shorthand (line) notation of the cell

(5 + 2 + 2 marks)

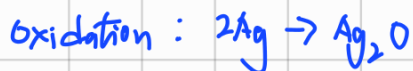
4. Which metal cation is the best oxidizing agent?

- i.  $Pb^{2+}$   
 ii.  $Cr^{3+}$   
 iii.  $Fe^{2+}$   
 iv.  $Sn^{2+}$

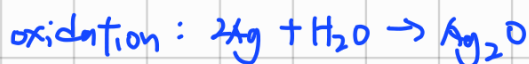
(1 mark)



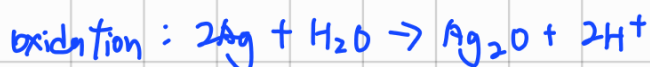
— Balance Ag



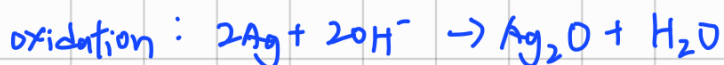
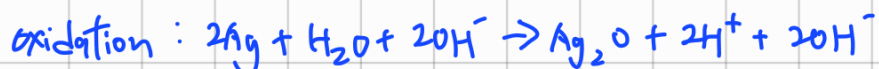
— Balance O



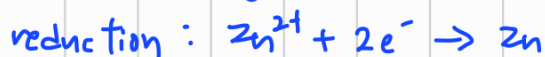
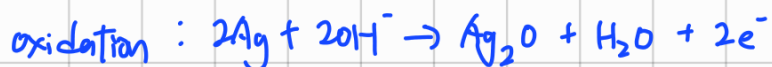
— Balance H



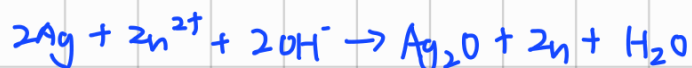
— Neutralise  $\text{H}^+$  with  $\text{OH}^-$

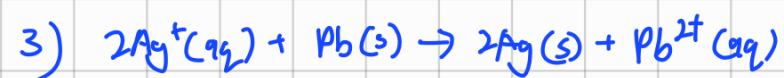


— Balance  $e^-$

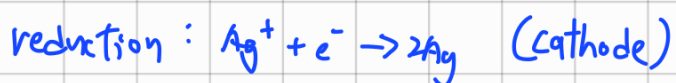
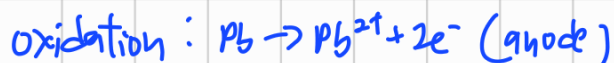
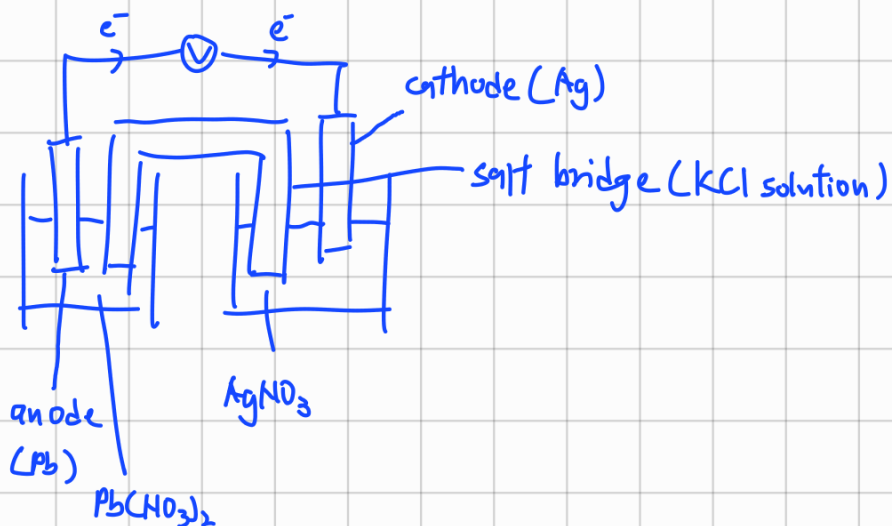


— Overall equation (Add up)





(i)



$$\begin{aligned} \text{(ii)} \quad E^\circ_{\text{cell}} &= E^\circ_{\text{cathode}} - E^\circ_{\text{anode}} \\ &= +0.80 - (-0.13) \\ &= +0.93 \text{ V} \end{aligned}$$



4)  $\text{Pb}^{2+}$  is the best oxidizing agent because its  $E^\circ$  value is most positive.

|       |       |    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |       |       |        |       |
|-------|-------|----|--|--|--|--|--|--|--|--|--|--|--|--|--|--|-------|-------|--------|-------|
| 1     |       |    |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 18    |       |        |       |
| IA    |       |    |  |  |  |  |  |  |  |  |  |  |  |  |  |  | VIIIA |       |        |       |
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| H     | 2     |    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |       | He    |        |       |
| 1.01  | IIA   |    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |       | VIIA  | 4.00   |       |
| 3     | 4     |    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |       | 10    |        |       |
| Li    | Be    |    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |       | Ne    |        |       |
| 6.94  | 9.01  |    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |       | F     | 19.00  | 20.18 |
| 11    | 12    |    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |       | 17    | 18     |       |
| Na    | Mg    |    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |       | Cl    | Ar     |       |
| 22.99 | 24.31 |    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |       | 35.45 | 39.95  |       |
| 19    | 20    |    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |       | 36    |        |       |
| K     | Ca    |    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |       | Kr    | 83.80  |       |
| 39.1  | 40.08 |    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |       | Br    | 79.90  |       |
| 37    | 38    |    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |       | 54    |        |       |
| Rb    | Sr    |    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |       | Xe    | 131.29 |       |
| 85.47 | 87.62 |    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |       | I     | 126.9  |       |
| 55    | 56    |    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |       | 53    | 86     |       |
| Cs    | Ba    |    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |       | At    | Rn     |       |
| 132.9 | 137.3 |    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |       | (210) | (222)  |       |
| 87    | 88    |    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |       | (209) |        |       |
| Fr    | Ra    |    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |       |       |        |       |
| (223) | (226) |    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |       |       |        |       |
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**LIST OF SELECTED CONSTANT VALUES**

|                            |                    |   |                                                                                                |
|----------------------------|--------------------|---|------------------------------------------------------------------------------------------------|
| Speed of light in a vacuum | $c$                | = | $3.0 \times 10^8 \text{ m s}^{-1}$                                                             |
| Avogadro's number          | $N_A$              | = | $6.02 \times 10^{23} \text{ mol}^{-1}$                                                         |
| Faraday constant           | $F$                | = | $9.65 \times 10^4 \text{ C mol}^{-1}$                                                          |
| Planck constant            | $h$                | = | $6.6256 \times 10^{-34} \text{ J s}$                                                           |
| Reduced Planck constant    | $\hbar$            | = | $1.054 \times 10^{-34} \text{ J s}$                                                            |
| Rydberg constant           | $R_H$              | = | $1.097 \times 10^7 \text{ m}^{-1}$<br>= $2.18 \times 10^{-14} \text{ J}$                       |
| Molar of gases constant    | $R$                | = | $8.314 \text{ J K}^{-1} \text{ mol}^{-1}$<br>= $0.08206 \text{ L atm mol}^{-1} \text{ K}^{-1}$ |
| Boltzmann constant         | $k$                | = | $1.3807 \times 10^{-27} \text{ J K}^{-1}$                                                      |
| Mass of proton             | $M_p$              | = | $1.672 \times 10^{-27} \text{ kg}$                                                             |
| Electronic Bohr magneton   | $m_B$              | = | $9.2741 \times 10^{-24} \text{ J T}^{-1}$                                                      |
| Nuclear Bohr magneton      | $b_N$              | = | $5.05 \times 10^{-27} \text{ J T}^{-1}$                                                        |
| Vapour pressure of water   | $P_{\text{water}}$ | = | 23.8 torr                                                                                      |
| Electron charge            | $e^-$              | = | $1.602 \times 10^{-19} \text{ C}$                                                              |

**UNIT AND CONVERSION FACTOR**

|          |                                                                                                                                                                                                                                          |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Energy   | $1 \text{ J} = 1 \text{ kg m}^2 \text{ s}^{-2} = 1 \text{ N m} = 10^7 \text{ erg}$<br>$1 \text{ calorie} = 4.184 \text{ Joule}$<br>$1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$<br>$1 \text{ amu} = 1.66 \times 10^{-27} \text{ kg}$ |
| Pressure | $1 \text{ atm} = 760 \text{ mm Hg} = 760 \text{ torr} = 101.325 \text{ kPa} = 101325 \text{ N m}^{-2}$                                                                                                                                   |

## STANDARD REDUCTION POTENTIAL ELECTROCHEMICAL SERIES

| Half-Reaction                                             | E (Volts) |
|-----------------------------------------------------------|-----------|
| $K^+ + e^- \rightleftharpoons K$                          | -2.924    |
| $Ba^{2+} + 2e^- \rightleftharpoons Ba$                    | -2.900    |
| $Ca^{2+} + 2e^- \rightleftharpoons Ca$                    | -2.760    |
| $Na^+ + e^- \rightleftharpoons Na$                        | -2.712    |
| $Mg^{2+} + 2e^- \rightleftharpoons Mg$                    | -2.375    |
| $H_2 + 2e^- \rightleftharpoons 2 H^-$                     | -2.230    |
| $Al^{3+} + 3e^- \rightleftharpoons Al$                    | -1.706    |
| $Mn^{2+} + 2e^- \rightleftharpoons Mn$                    | -1.040    |
| $Zn^{2+} + 2e^- \rightleftharpoons Zn$                    | -0.763    |
| $Cr^{3+} + 3e^- \rightleftharpoons Cr$                    | -0.740    |
| $S + 2e^- \rightleftharpoons S^{2-}$                      | -0.508    |
| $2CO_2 + 2H^+ + 2e^- \rightleftharpoons H_2C_2O_4$        | -0.490    |
| $Cr^{3+} + e^- \rightleftharpoons Cr^{2+}$                | -0.410    |
| $Fe^{2+} + 2e^- \rightleftharpoons Fe$                    | -0.409    |
| $Co^{2+} + 2e^- \rightleftharpoons Co$                    | -0.280    |
| $Ni^{2+} + 2e^- \rightleftharpoons Ni$                    | -0.230    |
| $Sn^{2+} + 2e^- \rightleftharpoons Sn$                    | -0.136    |
| $Pb^{2+} + 2e^- \rightleftharpoons Pb$                    | -0.126    |
| $Fe^{3+} + 3e^- \rightleftharpoons Fe$                    | -0.036    |
| $2H^+ + 2e^- \rightleftharpoons H_2$                      | 0.000     |
| $S_4O_6^{2-} + 2e^- \rightleftharpoons 2 S_2O_3^{2-}$     | 0.089     |
| $Sn^{4+} + 2e^- \rightleftharpoons Sn^{2+}$               | 0.150     |
| $Cu^{2+} + e^- \rightleftharpoons Cu^+$                   | 0.158     |
| $Cu^{2+} + 2e^- \rightleftharpoons Cu$                    | 0.340     |
| $O_2 + 2H_2O + 4e^- \rightleftharpoons 4OH^-$             | 0.401     |
| $Cu^+ + e^- \rightleftharpoons Cu$                        | 0.522     |
| $I_3^- + 2e^- \rightleftharpoons 3 I^-$                   | 0.534     |
| $MnO_4^- + 2H_2O + 3e^- \rightleftharpoons MnO_2 + 4OH^-$ | 0.588     |
| $O_2 + 2H^+ + 2e^- \rightleftharpoons H_2O_2$             | 0.682     |
| $Fe^{3+} + e^- \rightleftharpoons Fe^{2+}$                | 0.770     |
| $Hg_2^{2+} + 2e^- \rightleftharpoons Hg$                  | 0.796     |